

SLIIT
SE3082 – Parallel Computing



Lab 7
Exercise 01

Akash D
IT24101853

Exercise 01: Broadcast + Send/Recv

Implementation Code (sum_bcast.c)

```
#include <mpi.h>
#include <stdio.h>
#include <stdlib.h>
#define N 1000000

int main(int argc, char **argv) {
    MPI_Init(&argc, &argv);

    int rank, size;

    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &size);

    int *array = (int *)malloc(N * sizeof(int));

    if (rank == 0) {
        for (int i = 0; i < N; i++) array[i] = i + 1;
    }

    double start = MPI_Wtime();

    MPI_Bcast(array, N, MPI_INT, 0, MPI_COMM_WORLD);

    int chunk_size = N / size;

    int start_idx = rank * chunk_size;
    int end_idx = start_idx + chunk_size;

    long long local_sum = 0;

    for (int i = start_idx; i < end_idx; i++) local_sum += array[i];

    if (rank != 0) {
        MPI_Send(&local_sum, 1, MPI_LONG_LONG, 0, 0, MPI_COMM_WORLD);
    } else {
        long long total_sum = local_sum;

        for (int r = 1; r < size; r++) {
            long long recv_sum;

            MPI_Recv(&recv_sum, 1, MPI_LONG_LONG, r, 0, MPI_COMM_WORLD,
                MPI_STATUS_IGNORE);
```

```
total_sum += recv_sum;
}
double elapsed = MPI_Wtime() - start;
printf("\n[Ex1 Bcast] Total sum: %lld, Time: %.4f sec\n", total_sum, elapsed);
}
free(array);
MPI_Finalize();
return 0;
}
```

Test Output (4 processes)

[Ex1 Bcast] Total sum: 500000500000, Time: 0.0121 sec